A Probabilistic Instrument for Studying Reasoning Traces and Testing the Ordering and Markovian Assumption Behind Them

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Abstract

This paper builds a probabilistic instrument to analyze a chain of thought and find the parts across which the answer distribution changes. It types every sentence of a trace, groups the sentences into nodes at three levels of resolution and the nodes into blocks, resamples the model from the block boundaries, and estimates from the continuations the probability of the final answer and the type of the next move at each cut. Each large shift in the answer probability is a difference between two adjacent block boundaries. In the problem under study, a cognitive reflection item with its defining constraint deleted, put to Qwen3.6 27B, the instrument located the two blocks whose rise in the probability of the final answer, by 0.36 and by 0.60, survives every treatment of the unfinished continuations, and the moves inside the second show where the trace stops analyzing the problem and starts scoring answers. The Markovian assumption fails when the state is the type of the last move: cuts ending on the same kind of move, and even three cuts ending on the identical full label, give different distributions of the next move, so what the model does next depends on more of the prefix than the kind of move it just made. And the order of thoughts tracks the answer: in traces written from an empty prefix, those that work the algebra before naming the classic riddle behind the item end about eight times as often at the option that calls the question underdetermined but names the classic answer as intended, an exploratory association whose significance depends on how the unfinished traces are counted. The item is small by design, so that every quantity can be defined exactly and checked. The instrument applies wherever a long argument has a defined outcome at each cut and a budget for resampling: to reading long machine arguments such as the claimed Navier–Stokes proof of September 2026, to AI safety, where the end of a completed block of reasoning is a natural checkpoint before the reasoning goes on toward a harmful outcome, and to training, where the reasoning patterns it exposes can be weakened or strengthened.

1 Introduction

Reasoning models write long chains of thought before they answer (Wei et al., 2022; Guo et al., 2025), and that text is where much of their problem solving can be read. Reading it matters in three ways. For oversight, the chain of thought is the part of a model's process that can be inspected, though not a faithful transcript of it: models can be moved by features of the prompt that their reasoning never mentions (Turpin et al., 2023; Chen et al., 2025), some answers depend little on parts of the written reasoning (Lanham et al., 2023), some steps that look like reflection are decorative (Zhao et al., 2025), and the answer often settles well before the trace ends (Scalena et al., 2026; Boppana et al., 2026), so it is both a promising and a fragile window (Korbak et al., 2025), and a way to find the stretch of reasoning across which the outcome moves is a way to place a checkpoint before a harmful outcome is reached. For training, the same stretches name the reasoning patterns a model should be taught to avoid or to keep. And for the reader of a long machine argument, the question is which parts of the route carried the result. That last need is no longer hypothetical. In September 2026 an OpenAI system of some ten thousand cooperating agents produced, over 88 hours, a claimed resolution of the Navier–Stokes existence and smoothness problem, a book length writeup with a formalization in Lean (OpenAI, 2026; Castelvecchi, 2026). Whatever its final

assessment, a proof checker certifies that such an argument is correct without saying which parts of the route carried the result, which choices committed it to the construction it ended with, which known results entered and when, and which stretches were verification that changed nothing; for a mathematician or a physicist the route is often more instructive than the result, and no one reads a route of that length in full.

Both operations of the instrument are borrowed, with small additions that make them a little more adaptable and easier to use. Schemes for typing the moves of a trace are many (Lee et al., 2026; Gandhi et al., 2025; Venhoff et al., 2025; Marjanovic et al., 2026; Jiang et al., 2025; Li et al., 2025; Kargupta et al., 2025; Wang et al., 2025); ours takes the node types of ReasoningFlow (Lee et al., 2026) and adds a level at each end, blocks above, coarse units derived from the Level 1 nodes by fixed rules, and a third, item specific level below its subtypes, with the sentence kept whole as the unit. Resampling continuations from a prefix is the common way to read a trace, at every token (Bigelow et al., 2025), at sentence boundaries (Bogdan et al., 2025) or from truncated prefixes on the model's own distribution (Macar et al., 2026); these leave the placement of the cuts to the analyst, and the hierarchy offers one rule for it, not yet compared with others, since a change in the outcome need not sit at any one sentence: cut first at block boundaries, and go finer only where the outcome moved. Transitions between labels are usually counted over a corpus under a memoryless assumption (e.g., Shahariar et al., 2025), which we test by resampling from fixed prefixes; order has been studied in the prompt (Chen et al., 2024) and in finished chains (Chen et al., 2026), and answers have been traced to reasoning against retrieval (Wang et al., 2026), whereas our question is the order in which the model brings in what it knows while it writes. The item, the bat and ball problem of the cognitive reflection test (Frederick, 2005) with its defining clause deleted, joins an altered famous problem (Fascendini et al., 2026), a missing premise (Fan et al., 2025) and an ambiguous hedge on a task with a known intuitive error (Hagendorff et al., 2023). Beyond these additions, the paper contributes a registered test of whether the last label fixes the distribution of the next, with a comparison among cuts sharing an identical full label, and an exploratory measurement of the order of algebra and recognition of the classic riddle in traces written without a prefix.

This paper studies external reasoning, the reasoning a model carries out in the text it writes and then conditions on, as distinct from the internal reasoning of its activations, under one working assumption: that the two agree at a coarse level, a stretch of several sentences that completes one logical argument, a plan and the steps that follow from it, rather than a single sentence or token. The instrument localizes reasoning at that scale. It cuts a trace at block boundaries, resamples the model from each cut, and reads where the distribution of the outcome moves; a block across which it does not move can be skipped, a block across which it moves is where the consequential work was done, and the typed moves inside it say what kind of work that was, while two further questions ask whether what comes next depends only on the type of the last move and whether the order in which the model brings in what it knows is associated with where it ends. These are the questions one wants answered about a long machine argument, and the same instrument applies to a proof once each cut has a defined outcome and a block may be a lemma, at a cost that grows with the length. We demonstrate it at the smallest scale on which the questions can be asked: one model, Qwen3.6 27B in thinking mode (Section 3), on one item chosen because answering it calls for three kinds of move that can be told apart in a trace, algebra with the two prices, prior knowledge of the famous problem the item resembles, and the reading of the hedge words that turn two readings of "most likely" into two different letters; with the clause missing, any ball price below \$0.55 fits the text, so the answer is a matter of reading and the model's answers split between two options. Cuts sit at block boundaries because a change in the answer distribution need not sit at any one sentence: in an earlier sentence level study of the trace examined here (Maleki, 2026), the change inside one block was spread over several sentences (Section 2).

2 THE INSTRUMENT

The instrument reads a trace at several levels of resolution, beginning with its broad structure and introducing finer detail only where it is needed, in the way a physicist works through successive orders of approximation and gradient boosting adds models that correct the residuals of those before it, up to the round where refinement fits noise. We take from both the movement from coarse to fine, not their mathematics: the levels below do not form a perturbation expansion, and a finer level does not correct a coarser one. A representation that is too coarse may conceal the part of the reasoning

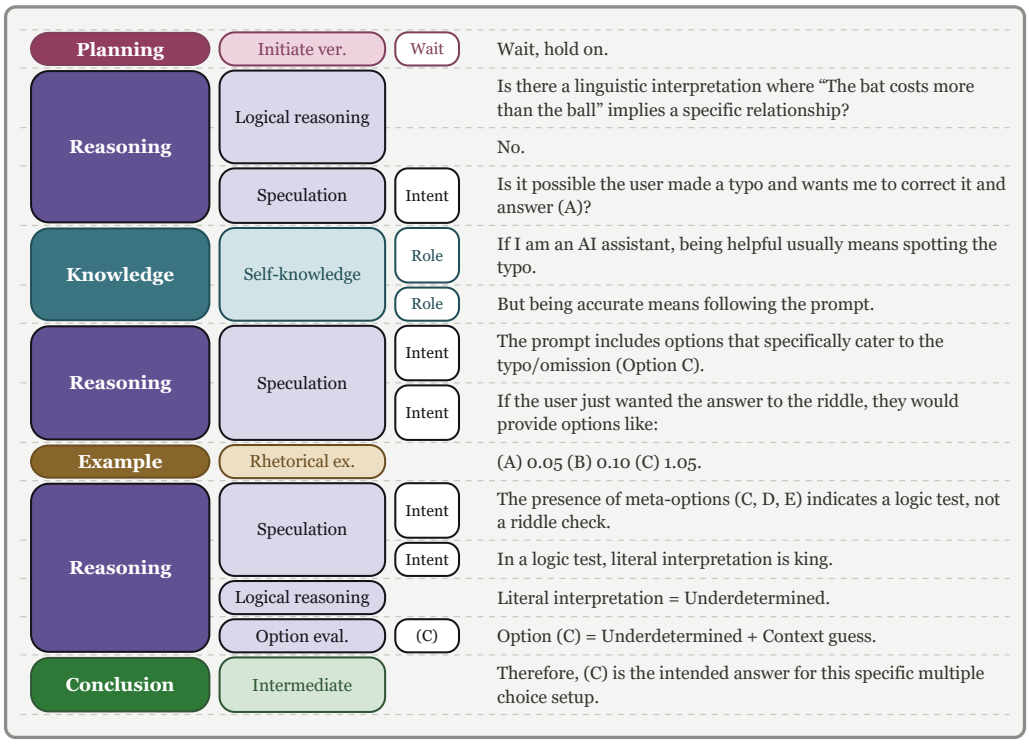


Figure 1: The instrument: one block of the C trace opened out. One row per sentence, the same span described at Level 1, Level 2 and Level 3, with the sentence itself at the right; a Level 3 box appears only where the label has a third level, and color marks the Level 1 kind.

that matters, whereas one that is too fine may introduce irrelevant fluctuations and depend more on arbitrary segmentation decisions. Figure 1 shows one block opened out to its sentences and their labels at the three levels.

Labels, nodes and blocks: At Level 1 a trace is a sequence of nodes of eight kinds, the node taxonomy of ReasoningFlow (Lee et al., 2026), whose node types we keep and whose edges we do not, with one change, the category Fact renamed Knowledge, because information a model brings into its reasoning need not be correct: Planning (Pl), Reasoning (Re), Restatement (Rs), Knowledge (Kn), Assumption (As), Reflection (Rf), Example (Ex) and Conclusion (Co). Level 1 is inherited and held fixed, which allows comparison with other traces and taxonomies. Level 2 divides each kind into the subtypes of ReasoningFlow and adds leaves specific to the item, such as evaluating an answer option or weighing a hedge word, and Level 3 exists only where a Level 2 leaf needs a further distinction; these finer levels may be defined for the question at hand. A label is thus a path in a tree, and a node at any level is a maximal run of consecutive sentences that share the label at that level, so a Level 1 node splits into Level 2 nodes, and those into Level 3 nodes, where the finer labels differ. Sentences are the finest level of the instrument: each carries one path, and we do not subdivide it. A sentence that does two things carries both paths, forms a node by itself, and counts under the first of its listed paths wherever a single label is needed, which for a sentence opened by "Wait" is the verification path. Appendix B summarizes the tree; its 77 label paths, with a mapping of the item leaves to general ones, are in the repository.

Blocks are built from the Level 1 nodes by four opener rules: a sentence opens a block if it is the first sentence of a Planning node that is not entirely a local plan, such as a step label inside a derivation; if it opens a case (Assumption, branching); if it gives the final answer (Conclusion, final answer); or if it begins with the word "Wait", alone or after "But", "And", "Oh" or "Okay". A block runs from one opener to the next and carries no type of its own. Given the labels the construction is deterministic, and the "Wait" rule is applied to the text by code; the labels are judgments, and the rules are a candidate segmentation, written on the two traces of this study and not yet compared with

other boundaries. Since the labels are assigned to sentences, the construction runs from the bottom up while the analysis reads from the top down; Figure 1 shows one block opened out.

Cuts and resampling: Let $B_0, B_1, \ldots$ be the blocks of a trace in order. The cut $\mathrm{cut}(B_m)$ is the text of the trace from its first sentence through the last sentence of B_m. At each cut the prefix is placed in the model's thinking channel as its own unfinished thinking and the model continues it; we draw $n = 25$ continuations per cut, and Figure 2 shows the cuts of both traces. Two reference conditions sit below the first block: cut 0, where the model thinks from an empty prefix, written $\mathrm{cut}(\emptyset)$ with estimate $\hat{P}_\emptyset$, and a condition without thinking, where the thinking block is closed empty and the model answers at once.

A continuation is read only if it finished: the model stopped of its own accord and closed its thinking, and its answer is the option letter on the last nonblank line of its reply. A continuation that reaches the limit of 16,000 generated tokens has no answer; we call it unresolved, write it "?", and keep it in every denominator. The quantity of interest at a cut is the probability

$$P_m(a) = P\big(\text{completed answer} = a \text{ within } 16{,}000 \text{ generated tokens} \;\big|\; \mathrm{cut}(B_m)\big), \tag{1}$$

a property of the model and the prefix that we never observe. What we report is the estimate $\hat{P}_m(a)$, the fraction of the 25 sampled continuations that give a, with intervals for its sampling uncertainty. P_m describes what the model outputs from a given prefix; it is not the answer the model would give with an unlimited budget and does not say when the model has settled internally, and the condition is always the whole prefix through B_m, never the block alone.

With a_T the trace's own final answer, the shift across block B_m is

$$\Delta_m = \hat{P}_m(a_T) - \hat{P}_{m-1}(a_T), \qquad \Delta_0 = \hat{P}_0(a_T) - \hat{P}_\emptyset(a_T). \tag{2}$$

Because the prefixes on either side differ only by the text of B_m, the shift is the difference between the output distributions conditioned on the two prefixes. Where inside the block the change arises is not measured, and since every continuation receives the same 16,000 new tokens, a shift can reflect information in the added text, work that text has already done, or a higher chance of finishing within the budget. We say a shift is localized between two block boundaries when the change across them is large and the changes across the neighbouring boundaries are not. Cuts are sampled in order, and a trace stops at the first cut where all 25 continuations give its answer; that is a criterion for stopping the sampling, not evidence that the answer can no longer change, and the later blocks are assumed, not measured, to leave it unchanged. The length of a span of text is $T_k(\cdot)$, the number of the model's own tokens in it, so that $T_k(B_m)$ is the length of a block and T_k of a node or a whole trace is defined the same way; block lengths are computed as the difference between the prompt token counts that the endpoint reports for consecutive prefixes.

The block is the unit of the cuts because a change in the answer distribution need not sit at any one sentence, and a block is coarse enough to hold such a change whole between two cuts; the evidence, from an earlier sentence level study of the C trace, opens the results (Section 4). The resolution of the intervention and the resolution of the interpretation are therefore separate: cuts at block boundaries identify the regions from which continuations reach different answers, and the labels inside a region say which moves are there and where knowledge enters, which is how Section 4 reads the blocks that move the answer; a move associated with an outcome is not thereby shown to have caused it. If the block level does not answer a question, the same resampling applies at node boundaries inside a block, at the cost of more variability.

The two trees: The same spine of cuts carries two trees. In the answer tree each cut branches into the answers its continuations reach, measured by $\hat{P}_m$ and Δ_m. In the transition tree each cut branches into the kind of move its continuations make first: every continuation is labeled by the judge described below, and the Level 1 label of the node that contains its first sentence is recorded; when that sentence continues the last node of the prefix, the node spans the cut and the first move repeats the last label. They are trees only in that they fan out from one spine of cuts, and they claim nothing about the topology of reasoning. Writing $W \to X$ for a move of kind W immediately before one of kind X and $W \dashrightarrow X$ for W somewhere before X, the quantities read from the

transition tree and from finished texts are

$$P(\text{next} = Y \mid \text{cut}(B_m)), \quad \text{the first move after cut } m; \tag{3}$$

$$P(\text{next} = Y \mid W \dashrightarrow X), \quad \text{given } W \text{ somewhere before the last move } X; \tag{4}$$

$$P(\text{next} = Y \mid W \to X), \quad \text{given } W \text{ immediately before } X; \tag{5}$$

$$P_{\text{corpus}}(Y \mid X), \quad \text{a node labeled } X \text{ followed by one labeled } Y \text{ in a corpus;} \tag{6}$$

$$P(a \mid \langle\, X \dashrightarrow Y \,\rangle), \quad \text{the answer, given a move of kind } X \text{ before one of kind } Y. \tag{7}$$

Equation 6 involves no resampling and no matching to block ends; Equation 7 is read from finished texts. Transitions are read at Level 1 unless stated otherwise; estimates of these quantities from the sampled continuations carry a hat, as $\hat{P}_m$ does, and the abbreviations above stand for the kinds of move.

The transition tree tests one structural assumption. Let X_m be the Level 1 label of the last node before cut m, $\hat{p}_m$ the distribution of the first moves of its labeled continuations, and $\bar{p}_X$ the distribution pooled over all continuations of the cuts with $X_m = X$. The statistic is the unweighted mean total variation distance

$$S = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \tfrac{1}{2} \left\| \hat{p}_m - \bar{p}_{X_m} \right\|_1, \tag{8}$$

where $\mathcal{M}$ holds the cuts whose last label is shared with at least one other cut. Under the null the first moves are exchangeable among the cuts that share a last label, so we permute the continuations among those cuts, keeping each cut's number of labeled continuations, 10,000 times, and report

$$p = \frac{1 + N_{\geq}}{10{,}001}, \tag{9}$$

where $N_{\geq}$ is the number of the 10,000 permutations whose statistic is at least the observed S. Continuations the judge could not label are left out of both the observed and the permuted statistics. Rejecting this null rejects one description of the state, the last Level 1 label, on the prefixes tested.

The two traces, and the judge: The traces come from an earlier sample of the model on the same prompt: of 50 requests, 44 returned a trace, 39 ending at E, 4 at C and one at the token limit, and six returned an error. No trace ended at any other option, so E and C were the only answers given. We took one complete trace for each of the two outcomes. The E trace was drawn at random; the C trace is the shortest of the four C traces, all of which were long (18,690 characters against 30,275, 45,162 and 47,993 for the other three). They were labeled in two steps: an LLM assistant labeled every sentence under the scheme, and we then reviewed and corrected the labels sentence by sentence. Under this reviewed labeling the C trace has 375 sentences and 47 blocks and the E trace 254 sentences and 38 blocks, about four Level 1 nodes each, and 9 sentences of the C trace and 5 of the E trace carry two labels.

The reviewed labeling is the reference: nodes and blocks are derived from it by code and the cuts sit at its block boundaries, so no judge places a cut. The continuations, and the 44 earlier traces, are labeled by an LLM judge, Claude Sonnet 5 with a prompt generated from the scheme, adaptive thinking at low effort and a retry on an invalid reply, which returns sentence labels only; nodes and blocks are again derived by code. Appendix C gives the selection of this configuration. Measured twice against the reviewed labels of the 629 source sentences, a week apart, the chosen configuration agrees with them at Level 1 with κ of 0.81 and 0.76, on the full label path with 0.70 and 0.68, on block boundaries with an F_1 of 0.92 and 0.93, and on the first node after each cut at 0.93 and 0.92; the gap between the two measurements is of the order of the judge's run to run variability, since two identical runs agree with each other at Level 1 with $\kappa = 0.90$ and on the full label path with 0.83. Relabeling the continuations of two cuts of the run gave a Level 1 κ of 0.78 and 0.83. Every number that depends on the judge carries these figures, while $\hat{P}_m$, Δ_m and the two reference conditions do not depend on the judge at all.

3 SETUP

Model and serving. The model is Qwen3.6 27B, an open weights reasoning model released by the Qwen team in April 2026 (repository `Qwen/Qwen3.6-27B`), in thinking mode, chosen because its thinking channel is exposed and can be continued from raw text. Continuations are drawn

from the raw completions endpoint of one provider (DeepInfra) with the chat template written out (Appendix F), so that the model continues the prefix as its own unfinished thinking, with no system prompt, temperature 1.0, the stop sequence <|im_end|>, a limit of 16,000 generated tokens and the provider's defaults otherwise.

The item. Every call uses the same prompt, given verbatim in Appendix F: the bat and ball question without its constraint, six options, (A) it is \$0.05, (B) it is probably \$0.05, (C) not enough information but most likely \$0.05, (D) I am not sure, (E) the question is not well posed and (F) none of the above, and an instruction to end the reply with "Answer: X". E says the question is broken; C says it is underdetermined but that \$0.05, the answer of the classic riddle, is the intended one; A states \$0.05 flatly. We grade no answer as correct or incorrect: the study measures which option the model selects, chiefly C against E, and how that choice changes with the prefix.

The run. Two reference conditions were sampled first, the model answering with an empty, closed thinking block (100 samples, up to 8,000 tokens, none reaching the limit) and cut 0, $\mathrm{cut}(\emptyset)$ (25 samples); the cuts of the two traces followed in order under the stopping rule, 12 for the C trace and 8 for the E trace, 625 continuations in all, of which the 525 with thinking were labeled by the judge, two remaining unlabeled after the retries and excluded from the transition analyses only. The questions, expectations, sample sizes, stopping rule, reading of answers and analyses were registered before the run; the registration, its three departures, one of which corrects the reading of answers, and the three independent computations of the headline quantities are in Appendix D. The stopping rule was applied with the registered scorer, under which the C trace first reached 25 of 25 at block 11; under the corrected reading used for every number in this paper the condition was met at block 9, so blocks 10 and 11 are observations past the stop. The registered optional second stage, node level cuts inside the block with the largest shift, was not run.

4 Results

The answer distribution across block boundaries: The block is the unit of the cuts for a reason found in the data. An earlier sentence level study of the same C trace (Maleki, 2026) placed ten cuts inside what is now its block 9; across them the estimated probability of C rose from 0.48 to 1.00 gradually though not monotonically, with a three point moving average that never falls and a largest step between neighbouring cuts of $+0.28$, below the jump of 0.4 that study had registered (Appendix D). Those estimates come from different cuts and an earlier scoring rule, so they illustrate rather than measure, but the change they show is spread over several sentences rather than carried by one.

Without thinking the model answered C 49 times in 100, A 32 and E 19; thinking from the empty prefix it answered E 18 times in 25 and C 3, with 4 unresolved. With the reasoning written out, the most frequent answer is E rather than C and A does not occur; the two conditions also differ in their token limits, 8,000 and 16,000, though no reply without thinking reached its limit.

In the C trace $\hat{P}_{\emptyset}(\mathrm{C}) = 0.12$ and $\hat{P}_{9}(\mathrm{C}) = 1.00$, a total movement of 0.88 (Figure 2a; Table 5). The largest shift, $\Delta_9 = +0.60$ $[0.37, 0.77]$, lies across block 9, sentences 58 to 78 under the heading "5. Evaluate the Options against the Textual Analysis:", with $T_k(B_9) = 321$ tokens or 6.8% of the trace. Two shifts of $+0.36$ follow it in size, Δ_0 across the opening restatement of the question ($T_k(B_0) = 139$) and Δ_6 across the block where the model recognizes the classic riddle ($T_k(B_6) = 87$). The path between them is a climb with setbacks: the three largest shifts sum to 1.32, more than the total movement, and one fall, $\Delta_3 = -0.28$ ($T_k(B_3) = 91$), has an interval that excludes zero before any adjustment for the twelve comparisons. The two interior rises are flanked by blocks that do not rise, Δ_9 by $\Delta_8 = \Delta_{10} = 0.00$ and Δ_6 by $\Delta_5 = +0.08$ and $\Delta_7 = -0.20$, and the opening rise Δ_0 is followed by $\Delta_1 = -0.20$.

The unresolved continuations qualify this reading: before block 9 between 6 and 12 of every 25 continuations reach the token limit, and from block 9 on none do. Assigning them in the least and the most favorable way within the observed batches, Δ_9 stays between $+0.32$ and $+0.60$ and Δ_0 between $+0.20$ and $+0.84$, while Δ_6 ranges from -0.08 to $+0.68$ and is left open (Table 7). The difference $\Delta_9 - \Delta_0 = 0.24$, with a bootstrap interval of 0.00 to 0.40, cannot separate block 9 from block 0 at 25 continuations per cut, and the registered rule for near ties names both. We had expected

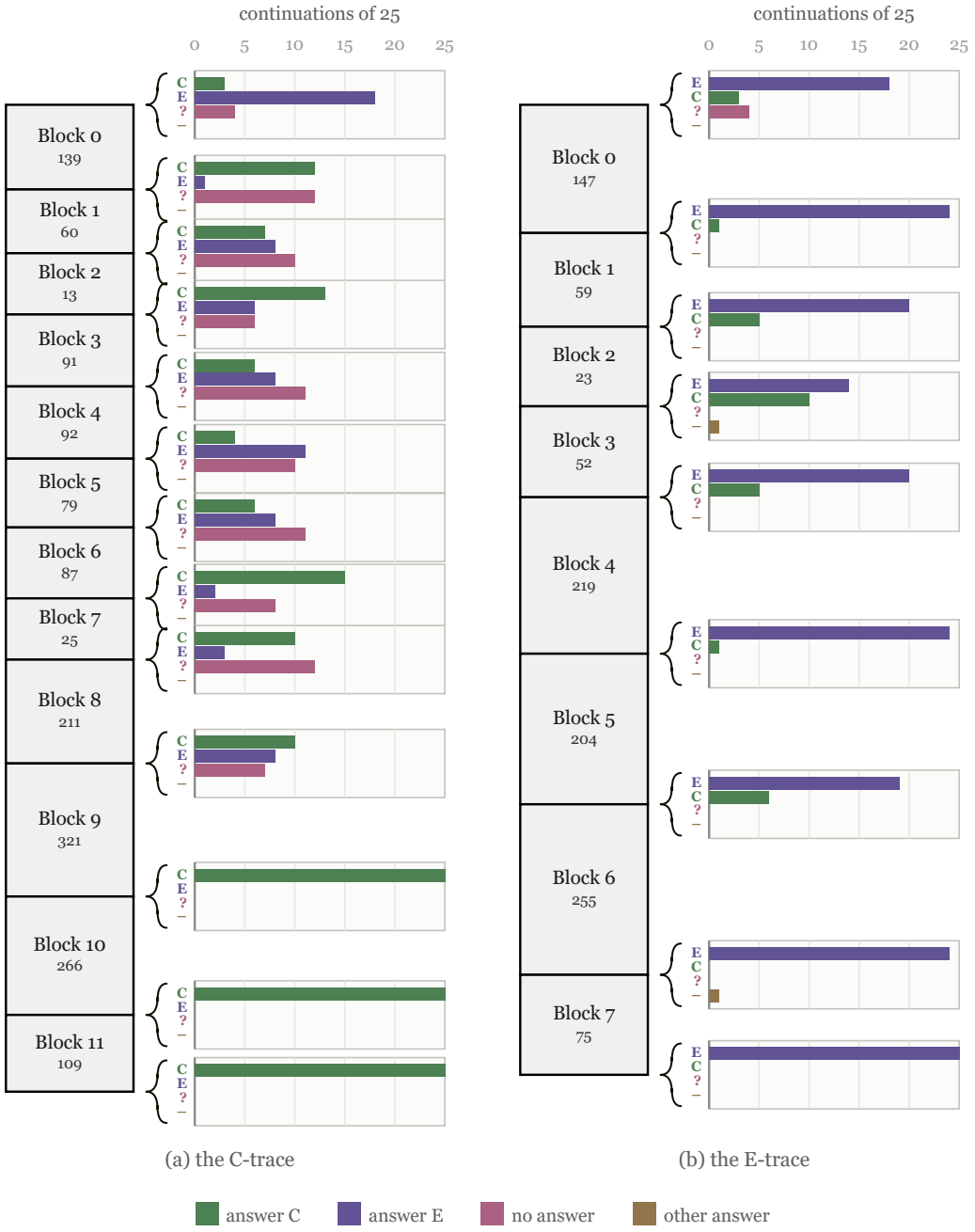


Figure 2: Where the answer moves, block by block. Each block of the trace is a box in the column; a cut is the boundary between two blocks, and the brace at each boundary opens onto the 25 continuations resampled from that cut, one bar per answer. A block's height is proportional to $M + \alpha T_k$, with $T_k(B_m)$ its length in the model's own tokens, so a short block stays legible beside a long one ($M = 90$, $\alpha = 0.5$; the number in each box is its T_k). Only the sampled blocks are drawn, 12 of the 47 blocks of the C trace and 8 of the 38 of the E trace; the uppermost cluster of bars in each panel is cut 0, the empty prefix, shared by the two traces.

Table 1: The registered test of the last label null: S observed against its mean under 10,000 permutations. Cuts whose last label occurs once are excluded; the pooled row regroups all 20 cuts.

trace	cuts	groups	S observed	S under the null	p
C trace	10	3	0.351	0.100	≤ 0.0001
E trace	4	2	0.090	0.040	0.0033
both	17	4	0.329	0.091	≤ 0.0001

the movement to spread over several blocks rather than one; two blocks carry the shifts that survive every assignment, with setbacks between them.

In the E trace $\hat{P}_{\emptyset}(\mathrm{E}) = 0.72$ and $\hat{P}_{7}(\mathrm{E}) = 1.00$, a total movement of 0.28 (Figure 2b; Table 6). $\hat{P}_{0}(\mathrm{E}) = 0.96$ after the opening block ($T_k(B_0) = 147$); it falls to $\hat{P}_{2}(\mathrm{E}) = 0.56$ across block 1 ($T_k(B_1) = 59$), which recognizes the classic riddle and states its standard answer, and block 2 ($T_k(B_2) = 23$), which writes the classic constraint into the algebra ("Ball = x, Bat = x + 1.00"); and $\hat{P}_{3}(\mathrm{E}) = 0.80$ after block 3 ($T_k(B_3) = 52$), a single sentence that notes the difference ("(Wait, the classic problem says…"). All 200 continuations of the E trace finished.

Inside the blocks that moved the answer: Block 9 of the C trace holds 21 sentences that take the options one at a time, with a small plan and a verdict for each. Sentences 65 to 67 are the trace's first analysis of a hedge word (" 'Probably' implies probability." "Unless there's a distribution of prices given, we can't assign probability."). Options A, B and C name the same price and differ only in how firmly they claim it, so scoring them is reading those words, and in terms of the labels block 9 is the first place where the trace stops analyzing the problem and starts scoring answers. The hedge word alone is not a rise: the E trace analyzes hedge words too, first in its block 5 ($T_k(B_5) = 204$), where $\Delta_5 = -0.20$.

At $\mathrm{cut}(B_8)$ ($T_k(B_8) = 211$), $\hat{P}_{8}(\mathrm{C}) = 0.40$, with 8 of the 25 continuations at E and 7 unresolved, yet $\hat{P}(\text{next} = \mathrm{Pl} \mid \mathrm{cut}(B_8)) = 24/24$: every labeled continuation opens a new block with Pl, as the original does. At $\mathrm{cut}(B_9)$, $\hat{P}(\text{next} = \mathrm{Pl} \mid \mathrm{cut}(B_9)) = 25/25$ again, and $\hat{P}_{9}(\mathrm{C}) = 1.00$. The first move is the same at both cuts while the answers differ sharply.

The two opening blocks restate the same material in different shapes, $\mathrm{Pl}(3) \to \mathrm{Rs}(5)$ in the C trace and $\mathrm{Pl}(4) \to \mathrm{Rs}(3) \to \mathrm{Pl} \to \mathrm{Rs} \to \mathrm{Pl} \to \mathrm{Rs}(2)$ in the E trace, the number in parentheses counting sentences; from an empty prefix 20 of 25 continuations open with the E trace's shape and only 2 like the C trace.

Does the last label fix what comes next? Across the 20 sampled cuts the first move opened a new block 73% of the time and reproduced the original trace's own next move 74% of the time.

The registered test rejects the null that the Level 1 label of the last move alone fixes the distribution of the next (Table 1). In the C trace the observed S is three and a half times the mean of its permutation null, and none of the 10,000 permutations reached or exceeded it, so $p = 1/10{,}001$; the E trace shows the same direction on four cuts, at a ratio of 2.3. We had expected the null to fail, and it does.

A Level 1 label covers unlike moves, so a finer state might absorb part of this spread. The comparison that survives the finest typing available is the following. Cuts 3, 4 and 5 of the C trace end on the identical full label path, the "Result:" sentence of successive cases in the enumeration "If Ball = \$0.05 / \$0.01 / \$0.55", and $\hat{P}(\text{next} = \mathrm{As} \mid \mathrm{cut}(B_m))$, the fraction that opens a new case, is $13/24$, $24/25$ and $11/25$ for $m = 3, 4, 5$ (an exploratory permutation test on the three cuts gives $S = 0.209$ against a null mean of 0.078, $p = 0.0006$). What differs between the three cuts is how many cases have already been tried, which in a single trace cannot be separated from position in the text or from the content of the prefix; a richer state, with finer labels, content or progress through the task, may well have Markovian dynamics.

An unmatched corpus reference over the 44 earlier traces and a comparison by history are given in Appendix E; both are confounded by position, since the cuts sit at block ends, where openers such as Planning are favored by construction, and within one trace earlier in the prefix means earlier in the text.

Table 2: Answer by the order of the judge's algebra leaf (A) and famous problem leaf (R) in the 69 texts written from an empty prefix (exploratory). Undefined texts never receive the famous problem leaf; unresolved texts stay in the denominator; intervals are Wilson 95%.

order	texts	C	E	unresolved	$\hat{P}(\mathrm{C})$ [95%]
$\langle A \dashrightarrow R \rangle$, algebra first	13	4	7	2	0.31 [0.13, 0.58]
$\langle R \dashrightarrow A \rangle$, riddle named first	52	2	47	3	0.04 [0.01, 0.13]
undefined, no famous problem leaf	4	1	3	0	

The order of algebra and recognition: The two source traces differ in the order of two moves the judge labels: algebra, the leaf on a sentence that sets up or manipulates the item's quantities, and the famous problem, the leaf on a sentence that names the classic riddle, its standard text or its standard answer. Under the reviewed labels the C trace does its algebra at sentence 10 and names the riddle at sentence 38; the E trace names it at sentence 14 and starts its algebra at sentence 17. We asked whether this order is associated with the answer on texts with no prefix, so that the order is the model's own: the 25 continuations of $\mathrm{cut}(\emptyset)$ and the 44 earlier traces, 69 texts labeled by the same judge. In each text A is the first sentence carrying the algebra leaf and R the first carrying the famous problem leaf, both read from the labels alone; four texts never receive the famous problem leaf and have no order, and no text has both leaves on one sentence.

Counting unresolved texts as not ending at C, $\hat{P}(\mathrm{C} \mid \langle A \dashrightarrow R \rangle) = 0.31$ against $\hat{P}(\mathrm{C} \mid \langle R \dashrightarrow A \rangle) = 0.04$, a difference of $+0.27$ $[0.07, 0.54]$, and Fisher's exact test gives $p = 0.012$; with the four undefined texts counted in the riddle first arm $p = 0.020$, under the most adverse assignment of the five unfinished texts $p = 0.07$, and among the texts that finished the figures are 4 of 11 against 2 of 49, $p = 0.008$ (Table 2). C is rare under both orders but about eight times as frequent when the algebra comes first; seven of the thirteen algebra first texts still end at E and two riddle first texts end at C, so the order neither fixes the answer nor is necessary for C. The direction holds in each batch separately, 1 of 4 against 1 of 19 at cut 0 and 3 of 9 against 1 of 33 in the earlier traces, though the 25 continuations alone cannot show it ($p = 0.32$).

In 52 of the 65 texts with an order the riddle is named before any algebra, in most of them at about sentence 14, right after the prompt is restated: recognition is the model's usual first move, and the association concerns the minority of texts that work the problem before placing it. The famous problem leaf marks recognition, not the use of the riddle's constraint; the scheme's leaf for assuming the missing "$1.00 more" premise appears in only 10 of the 69 texts, too few to test, so when the constraint's content is brought to bear remains unmeasured. The model chose both the order and the answer, nothing was manipulated, and an orientation toward one reading of the item could produce both. The analysis was added after registration, and two earlier versions of it, a hand reading of the texts and a rule applied to their words, are recorded and superseded in Appendix D.

5 Conclusion

We built an instrument that reads a reasoning trace at several resolutions: sentences are typed, nodes and blocks follow from the types by fixed rules, and the model is resampled at block boundaries to give two trees over one trace, of answers and of next moves. On one item and one model, the answer distribution moves in a few large shifts, each between two adjacent block boundaries; the type of the last move does not fix what comes next, even when the full label path of our scheme is identical; and in traces written without a prefix, the order in which the model does its algebra and names the classic riddle is associated with its answer.

Each result is stated at the scope of a case study: one model, one item, two traces chosen for their outcomes, 25 continuations per cut, labels drafted by an LLM and reviewed by us, a judge whose agreement with the reference is measured but not perfect, and a fixed continuation budget with no matched restart control. Appendix A sets out these limitations and the work that would address each, from a confirmatory test of the order association on fresh traces to a labeling scheme with separate axes for what a sentence does and what it is about, cuts at node boundaries, and more items and models.

AI USE STATEMENT

In this work, we used generative AI tools as follows. The project, its main idea and its design are ours: we chose the question, the item and the model, defined the labeling scheme and the block rules, set the hypotheses, the sample sizes and the stopping rule, reviewed and corrected every label of the two source traces, decided what counts as a finding and what is withdrawn, and revised all of the text. Claude (Anthropic; Claude Fable 5.1 at the time of writing) was used for consultation on the design of the methodology and experiments, for implementing methods, for supporting data analysis and interpreting results, for retrieving and summarizing related literature, and for drafting and polishing text. Nothing Claude proposed took effect without our review, and we consulted it at every step of our own work; the consultations and the decisions taken throughout the project are documented in the project repository (`https://github.com/drkianmaleki/reasoning-trace-experiments`), with every superseded version kept in its archive, so that the whole history of the decisions is available. The LLM judge that labels sentences is part of the method and is described in Section 2. Verification of the headline quantities ran in two stages: ChatGPT (OpenAI) first wrote an independent script from a written procedure, without sight of the registered tables, and we then recomputed the same quantities ourselves with our own script; the registered pipeline and the two later computations agree. ChatGPT also gave external review and polished parts of the text. Additionally, we used generative AI tools for routine lookup of syntax and inline code completion. We have reviewed all AI assisted work and take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The project repository (`https://github.com/drkianmaleki/reasoning-trace-experiments`) contains the registration and its amendments; the log of departures; the labeling scheme with its opener rules; the judge prompt with its label codes (sha256 in Appendix C); the analysis scripts; the tables of every cut; and the raw continuations. Every record stores its exact prompt and prefix. Sampling used an open weights model through a public endpoint with the settings of Section 3.

ETHICS STATEMENT

The study involves no human subjects and no personal data. The item is a public puzzle, and the model is an open weights model used through a commercial endpoint. The findings concern one model on one item and should not be read as general claims about reasoning models.

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A Limitations and future work

Scope: This is a case study: one model, one item, two traces, and for the order analysis one prompt condition. The two traces were chosen because one ends at C and one at E; 39 of the 44 traces in their pool end at E, and the C trace is the shortest of the four that end at C. The options of the item were never varied, so the pattern may depend on their particular wording and order. More items, more models and a second prompt condition are the direct extensions, and the same fork appears in safety relevant items on which a model must choose between asking and acting, which we have not tested.

The measurement: Twenty five continuations per cut resolve only large shifts: the standard error of one proportion near one half is 0.10 and of a difference between two such proportions about 0.14, so near one half a single shift below about 0.28 lies within sampling noise, a rough scale that is finer for frequencies near 0 or 1. The largest shift was selected as the largest of twelve and the stopping rule is adaptive; none of this is adjusted for in the intervals. A continuation from a long prefix receives that prefix and 16,000 new tokens, so the total budget differs between cuts and we have no restart control at a matched budget (Bulut, 2026); adding one, and more continuations at the cuts that matter, would sharpen the answer tree. The registered second stage, cuts at node boundaries inside the blocks that moved the answer, was not run and is the natural next measurement.

The labels: The source labels were drafted by an LLM assistant and reviewed by us, not labeled blind by a second human, and the scheme was written on these same two traces, so the sensitivity of

the block boundaries, and of the cuts, to the labeling was not measured; a blind second labeling, and a comparison of the block rules with other segmentations at matched numbers of cuts, would test both. The transition results rest on an LLM judge whose agreement with the reference and whose repeatability are reported, but whose first move labels at the tested cuts no independent human has checked, and a sentence with two labels counts under the first, a convention not tested. The scheme also puts what a sentence does and what it is about on one axis: algebra is a leaf under five parents and the famous problem has a single leaf that does not record how it enters the reasoning; a version with separate function and object axes is planned.

The order result: The association between order and answer was added after registration, its events are the judge's algebra and famous problem leaves, so it measures when the riddle is named rather than when its constraint is used, and its significance depends on how five unfinished texts are counted. A confirmatory test on fresh traces, with the events fixed in advance and a leaf that marks where the constraint enters the algebra, is the next step.

B THE LABELING SCHEME

Each sentence carries a path in a tree of labels. Level 1 is the eight node types of ReasoningFlow with Fact renamed Knowledge (Table 3). Level 2 takes the subtypes of ReasoningFlow, adds an unspecified residue leaf under each Level 1 label, and adds leaves specific to the item: under Reasoning, option evaluation, comparison and hedge word analysis; under Knowledge, self knowledge; under Planning, announcing the output. Level 3 exists only where a Level 2 leaf needs a distinction for the item: algebra, the famous problem, the option letters A to F, the everyday and the mathematical reading of a hedge word, the intent of the question's writer, well posedness, the missing "$1.00 more", the model's role, and the "Wait" doubt marker. A sentence that does two things carries both labels; there are 9 such sentences in the C trace and 5 in the E trace, and no sentence in either trace needed the unspecified leaf.

Sentences. A trace is split into sentences by a fixed rule: a newline always ends a sentence; within a line, a sentence ends after a period, question mark, exclamation mark or colon followed by white space; and a comma that closes a segment containing a relation symbol ($=$, $<$, $>$) also ends one, except inside parentheses. A heading or a list marker on a line of its own is therefore a sentence. Sentences and blocks are numbered from 0, so block 9 of the C trace runs from sentence 58 to sentence 78. A sentence with two labels forms a node by itself and counts under the first of its listed paths wherever one label is needed.

Table 3: The eight Level 1 labels, with examples from the two traces.

label	covers	example
Planning	announces what comes next, from a global heading to a local step label	"Let's check the options"
Reasoning	deduction, calculation, speculation, evaluation of an option	"then Bat = 1.10 − 0.55 = $0.55"
Restatement	repeats the question or an earlier sentence	the question and its options restated
Knowledge	brings in something the context did not contain, true or not	"this is the classic cognitive reflection test question"
Assumption	sets up a case or a premise not asserted as true	"If Ball = $0.05,"
Reflection	comments on earlier steps	"That checks out"
Example	a specific instance of something general	"(A) 0.05 (B) 0.10 (C) 1.05."
Conclusion	an intermediate or a final answer	"Therefore, (C) is the most logical choice."

Openers and blocks. A sentence opens a block if at least one of four rules holds. R1: it is the first sentence of a Planning node, and not every sentence of that node is a local plan. R2: it carries Assumption, branching, that is, it opens a case ("If Ball = $0.05,"). R3: it carries Conclusion, final answer. R4: it begins with the word "Wait", alone or after "But", "And", "Oh" or "Okay", once leading white space, list markers, markdown and parentheses are set aside, whatever its labels; such

a sentence is labeled as initiating verification or backtracking, with the Level 3 leaf for the doubt marker. Local plans never open a block: a step label with a colon inside a derivation ("Constraint:", "Substitute y:"), or a check question with its short answer ("Does Bat $>$ Ball?", "Yes."). An opener owns everything until the next opener, so a change of topic without a plan sentence stays inside its block, and nodes do not cross block boundaries. The C trace has 375 sentences, 193 Level 1 nodes and 47 blocks; the E trace has 254 sentences, 147 nodes and 38 blocks. Descriptively, the two traces build their blocks from the same openings and spend them differently: 162 of the C trace's sentences are Reasoning, and 57 of the E trace's are Restatement.

C THE JUDGE

Configuration. Claude Sonnet 5 with prompt v3, adaptive thinking at low effort, no sampling parameters and a retry on an invalid reply, run through the batch interface. Prompt v3 is generated by a script from the labeling scheme: 23,280 characters, about 5,820 tokens, 77 label paths, sha256 `e999f7081d5a285c4db1dcb4fb55f9eb5291b1ba536f770d1ad53a815e43f51a`; its full text is in the repository. The judge returns one line per run of consecutive sentences that share a label, as label codes and never as text, and a script expands the runs to sentences. Nodes and blocks are derived by code and never returned by the judge, and rule R4 is applied to the text by code.

Selection. Table 4 gives the measurements on the two reviewed traces, 629 sentences, against the reviewed labels. Prompt v2 was written from the E trace alone, with the C trace held out at the sentence level while it was written; the choice among configurations then used both traces. We also read a blind table of four columns, our own labeling and three Sonnet configurations, and ranked our own labeling first. The configuration chosen is the point of the curve at which the cost buys block structure: medium effort gains about as much as two identical runs differ and costs 1.6 times as much.

Table 4: Judge selection on the two reviewed traces. The two v2 low rows and the two v3 rows are recomputed against the current reviewed labeling (47 blocks); the other rows are from the original comparison, whose C trace then had 46 blocks, before rule R4 was added. The last column is the agreement on the first node after each reviewed cut.

model	prompt, thinking	Level 1 κ	block F_1	first node
Haiku 4.5	v1	0.29	C blocks collapse to 14	
Haiku 4.5	v2	0.34	C blocks collapse to 5	
Sonnet 5	v2, none	0.70	0.84	0.85
Sonnet 5	v2, low, two runs	0.72 to 0.74	0.92 to 0.93	0.89 to 0.90
Sonnet 5	v3, low, 2026-09-17	0.81	0.92	0.93
Sonnet 5	v3, low, 2026-09-24	0.76	0.93	0.92
Sonnet 5	v2, medium	0.79	0.93	0.95

The final prompt. The prompt used in the run adds to prompt v2 one sentence stating the "Wait" rule, whose boundary is enforced in code whatever the judge returns. Against the reviewed labels of the 629 source sentences it was measured twice at low effort, on 2026-09-17 and on 2026-09-24: pooled Level 1 κ 0.805 and 0.756, full path κ 0.699 and 0.675, block boundary F_1 0.920 and 0.933, first node agreement 0.928 and 0.916; the second run's Level 1 κ is 0.698 on the C trace and 0.820 on the E trace. The difference between the two measurements is of the order of the run to run variability below.

Repeatability. Two identical runs of the chosen configuration agree at Level 1 with $\kappa = 0.90$ and on the full label path with $\kappa = 0.83$; about one sentence in twelve changes its label between them. The block openers the judge misses recur at the same eight sentences in every Sonnet configuration and are ambiguities of the scheme. Relabeling the continuations of two cuts of the run gave a Level 1 κ of 0.78 and 0.83. Two continuations of 525 remained unlabeled after three attempts.

D PREREGISTRATION AND DEPARTURES

What was registered. The design was registered in a commit on 17 September 2026 at 14:52, before the run, together with amendments found by smoke tests before sampling began at 17:59, among them the stop sequence stated correctly and the token cap of the condition without thinking raised from 2,000 to 8,000 after seven of ten smoke replies were cut off. The registration fixes the object (the two traces and their 47 and 38 blocks), the model and endpoint, $n = 25$ per cut and 100 for the condition without thinking, the stopping rule, the scoring, the judge configuration with its prompt hash, the analyses of Section 2 with their statistics and numbers of permutations, and a rule for near ties that names the blocks for a second stage of cuts inside blocks, which was not run. The registered questions are how the probability of a trace's own answer moves across its blocks (Q1), whether the label of the next move depends only on the label of the last (Q2, a null to be tested), and how the resampled transitions compare with transitions counted over a corpus (Q3). Before the run we stated two expectations: that the movement of Q1 would spread over several blocks rather than one, and that the null of Q2 would fail. The pool of traces, the choice of the two traces and the earlier sentence level study came before the registration, and the order analysis of Section 4 came after it.

Departures. Three, all logged in the run folder. Two continuations of 525 could not be labeled after three attempts and are excluded from the transition analyses. The reading of answers was changed after the run: the registered scorer missed answers written "Answer: (C)" and, in continuations that never finished, read the first letter of ordinary words as an option ("Answer: Cannot be determined" became C). It misread 36 of 625 continuations. Under the registered reading $\Delta_9 = +0.56$, $\Delta_0 = +0.48$ and $\Delta_6 = +0.40$ in the C trace, $\hat{P}_{\emptyset}(\mathrm{C}) = 0.16$, the gap is 0.08 $[0.00, 0.32]$, and the first cut with 25 of 25 is block 11. The direction and the ranking of the three blocks are the same; the sizes differ because the registered reading credited answers to continuations that never finished. Under the corrected reading the stopping rule would have ended the C trace at block 9, so blocks 10 and 11 were sampled beyond it. The E trace is unaffected: all its continuations finished.

The earlier sentence level study. Ten of its cuts fall inside what is now block 9 of the C trace; in order, its estimated probability of C at them is 0.48, 0.60, 0.88, 0.88, 1.00, 0.88, 0.88, 1.00, 1.00 and 1.00, under that study's own cut positions and scoring rule.

Three computations. The proportions, shifts and intervals of Section 4 were computed three times from the same records by separate code: by the registered pipeline, by an independent script that a second AI system wrote from a written procedure without sight of the registered tables, and by a script of our own. All three agree on every count, proportion, shift and interval; they share one written procedure, so they verify the computation, not the sampling or the measurement decisions.

Superseded versions of the order analysis. Three earlier readings of the order question are on record. The first, of 2026-09-18, used the same two leaves but pooled the continuations of every cut of both traces, reporting $\hat{P}(\mathrm{C} \mid \text{algebra before the famous problem}) = 0.58$ over 293 continuations and $\hat{P}(\mathrm{E} \mid \text{famous problem before algebra}) = 0.81$ over 225; it was withdrawn because 289 of the 293 and 197 of the 225 came from cuts of one trace and inherited that trace's order from the prefix. The second, of 2026-09-19, read the texts by hand for the first algebra and the first import of the "$1.00 more" constraint and found 5 of 16 algebra first texts ending at C against 2 of 53 ($p = 0.006$); the third, of 2026-09-24, applied a rule to the words of the sentences for the same two events (`scripts/s7_order_prefixfree.py`) and found 5 of 19 against 2 of 50 ($p = 0.015$). Both were superseded on 2026-09-24 by the reading in Section 4, which takes the two events from the judge's labels as the rest of the paper does, on the 69 prefix free texts only, so that the instrument's own output is what is measured and no second reading of the text is introduced.

E FULL TABLES, AND THE CORPUS REFERENCE

Corpus and history comparisons. Over the 44 earlier traces, $P_{\text{corpus}}(\mathrm{Pl} \mid \mathrm{Re}) = 0.45$ and $P_{\text{corpus}}(\mathrm{Co} \mid \mathrm{Re}) = 0.25$. Pooled over the cuts with $X_m = \mathrm{Re}$, $\hat{P}(\text{next} = Y \mid \mathrm{cut}(B_m))$ is 0.47 for Pl, 0.26 for As, 0.25 for Re and 0.00 for Co, a total variation distance of 0.49 from the corpus row;

per cut the distance ranges from 0.22 to 0.97. Among the same cuts, $\hat{P}(\text{next} = Y \mid \text{Kn} \dashrightarrow \text{Re})$ differs from its counterpart for prefixes with no Knowledge move by a total variation distance of 0.655, and the same comparison for Rf and for Ex gives 0.574; where such a move occurred the next move is Pl in 70% to 81% of continuations, and where none did, As in 42% to 65%.

Table 5: The C trace. Counts of the 25 continuations at the cut after each block, the completed C frequency with its Wilson 95% interval, the shift with its Newcombe 95% interval, and the block's length $T_k(B_m)$ in the model's tokens (the trace is 4,740 tokens). $\text{cut}(\emptyset)$ gives C 3, E 18 and 4 unresolved. Other counts a completed answer other than C or E; ? counts unresolved continuations, drawn as no answer in the figures.

block	ends	C	E	other	?	$\hat{P}_m(\text{C})$ [95%]	Δ_m [95%]	$T_k(B_m)$
0	s7	12	1	0	12	0.48 [0.30, 0.67]	+0.36 [0.11, 0.56]	139
1	s12	7	8	0	10	0.28 [0.14, 0.48]	−0.20 [−0.43, 0.07]	60
2	s14	13	6	0	6	0.52 [0.33, 0.70]	+0.24 [−0.03, 0.47]	13
3	s22	6	8	0	11	0.24 [0.11, 0.43]	−0.28 [−0.50, −0.01]	91
4	s30	4	11	0	10	0.16 [0.06, 0.35]	−0.08 [−0.30, 0.14]	92
5	s35	6	8	0	11	0.24 [0.11, 0.43]	+0.08 [−0.14, 0.30]	79
6	s41	15	2	0	8	0.60 [0.41, 0.77]	+0.36 [0.09, 0.57]	87
7	s42	10	3	0	12	0.40 [0.23, 0.59]	−0.20 [−0.43, 0.07]	25
8	s57	10	8	0	7	0.40 [0.23, 0.59]	0.00 [−0.25, 0.25]	211
9	s78	25	0	0	0	1.00 [0.87, 1.00]	+0.60 [0.37, 0.77]	321
10	s97	25	0	0	0	1.00 [0.87, 1.00]	0.00 [−0.13, 0.13]	266
11	s106	25	0	0	0	1.00 [0.87, 1.00]	0.00 [−0.13, 0.13]	109

Table 6: The E trace, as in Table 5, with the completed E frequency. Every continuation finished. $\text{cut}(\emptyset)$ gives E 18, C 3 and 4 unresolved.

block	ends	C	E	other	?	$\hat{P}_m(\text{E})$ [95%]	Δ_m [95%]	$T_k(B_m)$
0	s11	1	24	0	0	0.96 [0.80, 0.99]	+0.24 [0.03, 0.44]	147
1	s15	5	20	0	0	0.80 [0.61, 0.91]	−0.16 [−0.35, 0.03]	59
2	s18	10	14	1	0	0.56 [0.37, 0.73]	−0.24 [−0.46, 0.02]	23
3	s19	5	20	0	0	0.80 [0.61, 0.91]	+0.24 [−0.02, 0.46]	52
4	s41	1	24	0	0	0.96 [0.80, 0.99]	+0.16 [−0.03, 0.35]	219
5	s52	6	19	0	0	0.76 [0.57, 0.89]	−0.20 [−0.40, 0.00]	204
6	s70	0	24	1	0	0.96 [0.80, 0.99]	+0.20 [0.00, 0.40]	255
7	s74	0	25	0	0	1.00 [0.87, 1.00]	+0.04 [−0.10, 0.20]	75

Table 7: Bounds on the three largest shifts of the C trace under any assignment of the unresolved continuations, holding the observed batches fixed. These are bounds on the missing outcomes, not sampling intervals.

block	before: C, ?	after: C, ?	observed shift	range
9	10, 7	25, 0	+0.60	+0.32 to +0.60
0	3, 4	12, 12	+0.36	+0.20 to +0.84
6	6, 11	15, 8	+0.36	−0.08 to +0.68

F THE PROMPT AND THE TEMPLATE

Every call uses the same prompt, shown here with its line breaks removed:

> A bat and a ball cost $1.10 in total. The bat costs more than the ball. How much does the ball cost? Choose the single best option: (A) It is $0.05 (B) It is probably $0.05 (C) There is not enough information, but the answer is most likely $0.05 (D) I am not sure (E) The question is not well posed (F) None of the above. Explain your reasoning, then end your reply with a single line of the form "Answer: X" where X is exactly one option letter.

Continuations are requested from the raw completions endpoint with the chat template written out, the prefix standing in the model's thinking channel:

```
<|im_start|>user\n{prompt}<|im_end|>\n
<|im_start|>assistant\n<think>\n{prefix}
```

We did not pin a model revision, and a chat endpoint tried first discarded prefills of the thinking channel, which is why the raw endpoint is used. The condition without thinking closes the thinking block empty, `<think>\n\n</think>\n\n`, and the model answers at once.